

looters in Kabul a few days after it was captured by Northern Alliance forces on November 13. The files provide information about reconnaissance missions to Europe and the Middle East. A report in the UK's Independent newspaper indicates that the encryption used to protect these files had been significantly weakened by US export restrictions that existed until last year. The files were reportedly stored using Microsoft's Windows 2000 operating system and protected from unauthorised access using the Encrypting File System (EFS), which comes as standard on this platform. They were protected with a 40-bit Data Encryption Standard (DES), according to the Independent's report.

This was the maximum strength encryption allowed for export by US law until March 2001. All systems are now sold with the standard 128-bit key encryption, exponentially stronger than 40-bit.

The Competitors

The computer-security market is rapidly coalescing around a few big players, including firewall leaders Check Point and NetScreen, and anti-virus company, Symantec. In this down economy, smaller companies that may have better technology, but less staying power are struggling to land clients. Often, it's hard for them to attract big customers who need assurance that a supplier will be around to support its product.

Competition is strong among the big name vendors. Cisco Systems has gained ground on several key server rivals, such as Juniper Networks, during this tech spending downturn. Cisco has been a big player in firewalls, intrusion-detection systems (IDS), and virtual private networks (VPNs) that use encryption to secure remote connections to corporate computer systems.

A VPN is a means by which certain authorised individuals (such as remote employees) have secure access to an organisation's intranet, by means of an extranet (a part of the internal network that is accessible via the Internet). VPN can be far less expensive than using actual private lines in a wide area network (WAN). VPNs can be strictly software, but most often they are hardware devices.

Customers would like to buy more than one security solution from a single company rather than piece together the products of several suppliers. Venture capitalists put \$1.6 billion into 140 computer-security startups during 2000. That fell 38 percent to \$990 million invested in 100 startups, during 2001. In 2002, some 60 startups received less than \$500 million through the first three quarters.

In 2001, corporations spent on an average 3 per cent of their tech budgets on security. Many analysts are expecting this market to grow to \$15 billion 2006 from what was \$6 billion in

Encryption provides a secure communication channel even when the underlying system and network infrastructure is not secure

2001. Over the next 10 years, markets for firewalls and VPNs, both relatively mature products, should grow at rates above 10 per cent.

On the leading edge, in areas such as wireless security, competition is especially fierce. Startups that build security options for cell phones and wireless local area networks are competing with established players such as Check Point and General Dynamics for a market that is still more promise than a reality.

Products Offered

Faster growth is expected from advanced forms of protection, such as intrusion detection, and tools that protect specific applications on corporate networks and vulnerability-assessment software and services. An Intrusion Detection System (IDS) inspects all inbound and outbound network activity and identifies suspicious patterns that may indicate a network or system attack from someone attempting to break into or compromise a system. This trend should continue for the next few years as more companies add IDS to their existing security infrastructure and the technology becomes easier to use.

Vulnerability assessment involves engaging tech experts to check a corporate network's security by probing and testing it. The market for this service was \$500 million in 2002, with sales of automated diagnostic tools probably making up 10 per cent of the total. Annual vulnerability checks by humans can cost as much as \$150 each. The same amount or less can buy multiple checks using automated systems.

Another growth area is protecting servers and desktop machines. Most companies, large and small, now use some form of perimeter firewall protection to protect their networks combined with antivirus software on mail servers and individual desktops. In fact, corporate antivirus software alone will be a \$2.7 billion market by 2006, up from \$1.2 billion in 2001.

From a security company's viewpoint, antivirus protection is not enough. Desktop firewalls are mandatory for many companies, leading to big deals for suppliers such as San Francisco-area companies ZoneLabs and Sygate. Both these companies have gotten big venture financing in recent months—proof that investors believe the market is ripe for these products.

The wireless LAN (WLAN) industry is experiencing tremendous growth aided by lower pricing points and standardisation of the technology. Analysts believe that this growth will continue to accelerate, all the multi-billion dollar technology and telecommunications companies such as Microsoft, Intel, Dell, AT&T, and IBM, to name a few, enter the WLAN industry. These corporations bring their deep financial pockets and millions of consumers to the WLAN industry. ■

Better Safe Than Sorry

As IT organisations face greater pressure to manage, store, and retrieve increasing amounts of corporate information, they are searching for new ways to keep data readily available, whenever and wherever needed. To support the requirement for Business Continuity and Availability (sometimes referred to as 'BC&A'), there are a wide variety of solutions available in the industry for various kinds of organisations. To name a few, cluster solutions, replication solutions and to some extent backup and recovery solutions qualify as BC&A solutions based on the requirement of the business.

For business requirements one solution which stands out and provides the most secure and robust mechanisms of data replication and availability of applications is storage-based replication. Based on the industry standard Fiber Channel (FC) protocol, this mechanism is the default choice for businesses today.

In a typical scenario, the infrastructure would comprise of servers, SANs, storage subsystems and IP networks for connectivity to the information islands. To achieve a true Disaster Recovery (DR) capability, the same and scaled down version of infrastructure has to be maintained at the DR site as well. So, there would be a set of servers in production using the production copy of data at the primary site and another set of servers at the DR/BC site waiting for the primary servers/site to fail. The servers on DR site can also be used in active-active configurations where these servers are performing some other computing tasks for the organisation.

To achieve Business Continuity and Disaster Recovery (BCP/DR), each one of these components has to be taken into consideration and robust methodologies applied to each component. Storage Area Networks and storage subsystems at the core of this infrastructure require special attention. For connectivity requirements between two sites

Effective Disaster Recovery practices are imperative for the effective functioning of any IT setup

refer to Figure 1 which states the technologies available and where they best fit. Other than the link between the sites, such technologies would essentially require SAN infrastructure, storage subsystems with licensed replication software (like PPRC, SRDF, True Copy, Continuous Access) and routers which would do long distance propagation and protocol conversion, if needed.

Long-distance SAN extension works on the storage-based replication of the storage subsystems and requires interconnects which can propagate the FC protocol over long distances. This propagation can be done in native FC protocol or by converting FC into IP and sending the signal over infinite distances. Although long distance FC connectivity is generally the most reliable, organisations sometimes prefer to send FC over existing long-distance IP links to take advantage of existing IP infrastructure.

It is essential to consider future needs when choosing the appropriate technology.

Today, the most common approach and also the most logical approach towards building a DR practice is to build a BC site not very

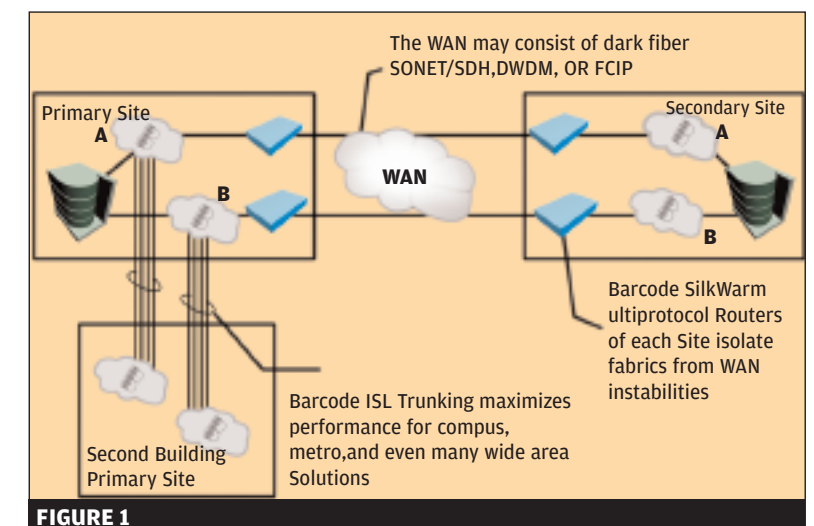


FIGURE 1